ABHYANKAR'S LEMMA: FORMAL SMOOTHNESS OF THE EXTENSION OF THE LOCAL RINGS AFTER ADJOINING ROOTS OF THE UNIFORMIZER

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Allowed standard prerequisites (editorial).

- The notation of the construction remark that names the extensions of the discrete valuation rings and the maximal ideals occurring in the statement. used in the statement at line 34041 and in the proof from line 34050. Fixes the objects A_1, B_1 and the maximal ideals m_i, m_ij so that the statement and the proof talk about the same rings.
- For an extension of discrete valuation rings, formal smoothness in the adic topology is detected by the ramification index being one and the residue field extension being separable. invoked at proof lines 34070-34072 and again at the end at proof lines 34133-34135. Reduces the two comparison steps to the computation of the ramification index and the separability of the residue field extension.
- The integral closure of a discrete valuation ring in the extension generated by a root of the uniformizer is a discrete valuation ring with ramification index the order of the root. invoked at proof lines 34092-34095. Identifies the intermediate ring A'_1 and its ramification index, which lets the argument pass to the subextension generated by the root.
- Formal smoothness goes up along the diagram of local extensions used to compare the intermediate ring with the original one. invoked at proof lines 34096-34102. Lets the conclusion for the subextension be transferred back to the extension appearing in the statement.
- Ramification indices of a tower of extensions of discrete valuation rings are multiplicative. invoked at proof lines 34124-34126. Computes the ramification index of the maximal ideal of B_1 over A_1 as one in the final case.

Source reference. Original remark 09EM: sources/stacks-more-algebra/pinned-original-source.tex, source lines 33882–33882. The statement at line 34041 uses the notation of the remark; the remark is a pointer that names the rings and ideals, and every argument in which those objects occur is inside the counted span.

Source reference. Original lemma 09E7: sources/stacks-more-algebra/pinned-original-source.tex, source lines 32989–32989. Used at proof lines 34070-34072 to reduce the first comparison to a computation of ramification index and residue field, and again

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at the very end at proof lines 34133-34135; both computations are carried out inside the counted span.

Source reference. Original lemma 09EV: sources/stacks-more-algebra/pinned-original-source.tex, source lines 33954–33954. Invoked at proof lines 34092-34095 to identify the intermediate ring and its ramification index; the verification that the exponent divides and that the root lies in the larger field is inside the counted span.

Source reference. Original lemma 09EQ: sources/stacks-more-algebra/pinned-original-source.tex, source lines 34006–34006. Applied at proof lines 34096-34102 to reduce the statement to the case of the subextension generated by the e-th root; constructing that subextension and the diagram it is applied to is inside the counted span.

Source reference. Original lemma 0BRL: sources/stacks-more-algebra/pinned-original-source.tex, source lines 32952–32952. Used at proof lines 34124-34126 to conclude that the ramification index of the maximal ideal is one; identifying the two extensions making up the tower and the separability of the residue field extension is inside the counted span.

Lemma 0.2 (Abhyankar's lemma)**.** *Let $A \subset B$ be an extension of discrete valuation rings. Assume that either the residue characteristic of A is 0 or it is p, the ramification index e is prime to p, and κ_B/κ_A is a separable field extension. Let K_1/K be a finite extension. Using the notation of Remark 09EM assume e divides the ramification index of $A \subset (A_1)_{\mathfrak{m}_i}$ for some i. Then $(A_1)_{\mathfrak{m}_i} \subset (B_1)_{\mathfrak{m}_{ij}}$ is formally smooth in the $\mathfrak{m}_{ij}$-adic topology for all $j = 1, \ldots, m_i$.*

Proof. Let $\pi \in A$ be a uniformizer. Let π_1 be a uniformizer of $(A_1)_{\mathfrak{m}_i}$. Write $\pi = u\pi_1^{e_1}$ with u a unit of $(A_1)_{\mathfrak{m}_i}$ and e_1 the ramification index of $A \subset (A_1)_{\mathfrak{m}_i}$.

Claim: we may assume that u is an eth power in K_1. Namely, let K_2 be an extension of K_1 obtained by adjoining a root of $x^e = u$; thus K_2 is a factor of $K_1[x]/(x^e - u)$. Then K_2/K_1 is a finite separable extension (by our assumption on e) and hence $A_1 \subset A_2$ is finite. Since $(A_1)_{\mathfrak{m}_i} \to (A_1)_{\mathfrak{m}_i}[x]/(x^e - u)$ is finite étale (as e is prime to the residue characteristic and u a unit) we conclude that $(A_2)_{\mathfrak{m}_i}$ is a factor of a finite étale extension of $(A_1)_{\mathfrak{m}_i}$ hence finite étale over $(A_1)_{\mathfrak{m}_i}$ itself. The same reasoning shows that $B_1 \subset B_2$ induces finite étale extensions $(B_1)_{\mathfrak{m}_{ij}} \subset (B_2)_{\mathfrak{m}_{ij}}$. Pick a maximal ideal $\mathfrak{m}'_{ij} \subset B_2$ lying over $\mathfrak{m}_{ij} \subset B_1$ (of course there may be more than one) and consider

$$\begin{array}{ccc} (B_1)_{\mathfrak{m}_{ij}} & \longrightarrow & (B_2)_{\mathfrak{m}'_{ij}} \\ \uparrow & & \uparrow \\ (A_1)_{\mathfrak{m}_i} & \longrightarrow & (A_2)_{\mathfrak{m}'_i} \end{array}$$

where $\mathfrak{m}'_i \subset A_2$ is the image. Now the horizontal arrows have ramification index 1 and induce finite separable residue field extensions. Thus, using the equivalence of Lemma 09E7, we see that it suffices to show that the right vertical arrow is formally smooth in the $\mathfrak{m}'_{ij}$-adic topology. Since u has a eth root in K_2 we obtain the claim.

Assume u has an eth root in K_1. Since $e|e_1$ and since u has a eth root in K_1 we see that $\pi = \theta^e$ for some $\theta \in K_1$. Let $K'_1 = K[\theta] \subset K_1$ be the subfield generated by θ. By Lemma 09EV the integral closure A'_1 of A in $K[\theta]$ is the discrete valuation ring $A'_1 = A[\theta]$ which has ramification index e over A. If we can prove the lemma for

the extension K_1'/K, then we conclude by Lemma 09EQ applied to the diagram

$$\begin{array}{ccc} (B_1')_{B_1' \cap \mathfrak{m}_{ij}} & \longrightarrow & (B_1)_{\mathfrak{m}_{ij}} \\ \uparrow & & \uparrow \\ A_1' & \longrightarrow & (A_1)_{\mathfrak{m}_i} \end{array}$$

for all $j = 1, \ldots, m_i$. This reduces us to the case discussed in the next paragraph.

Assume $K_1 = K[\pi^{1/e}]$ and set $\theta = \pi^{1/e}$. Let π_B be a uniformizer for B and write $\pi = w\pi_B^e$ for some unit w of B. Then we see that $L_1 = L \otimes_K K_1$ is obtained by adjoining θ/π_B which is an eth root of the unit w. Thus $B \subset B_1$ is finite étale. Thus for any maximal ideal $\mathfrak{m} \subset B_1$ consider the commutative diagram

$$\begin{array}{ccc} B & \xrightarrow[1]{} & (B_1)_{\mathfrak{m}} \\ {\scriptstyle e}\uparrow & & \uparrow{\scriptstyle e_{\mathfrak{m}}} \\ A & \xrightarrow{e} & A_1 \end{array}$$

Here the numbers along the arrows are the ramification indices. By multiplicativity of ramification indices (Lemma 0BRL) we conclude $e_{\mathfrak{m}} = 1$. Looking at the residue field extensions we find that $\kappa(\mathfrak{m})$ is a finite separable extension of κ_B which is separable over κ_A. Therefore $\kappa(\mathfrak{m})$ is separable over κ_A which is equal to the residue field of A_1 and we win by Lemma 09E7. □